

Insecurity and Blue Economy: An Appraisal of Rivers State

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Abstract

Insecurity remains a major challenge to the development of the blue economy in Rivers State. The state has extensive rivers, creeks, estuaries, and Atlantic coastlines that provide considerable opportunities for fishing, maritime transport, tourism, and other marine-based economic activities. This study, anchored in human security theory, assesses the relationship between insecurity and the development of the blue economy in Rivers State. In particular, it describes the historical background of the economy of Nigeria and discusses the prospects of the blue economy in Rivers State, as well as the security threats to its development and their socioeconomic and environmental consequences. The research method used in this study is the historical research method, whereby evidence was derived from oral interviews and documentary sources, such as books, journal articles, government publications, reports, newspapers, and other sources of information. The data were analysed qualitatively through a historical analysis. The results showed that piracy, kidnapping, illegal oil bunkering, cult-related violence, environmental degradation, poor maritime governance, and lack of adequate maritime infrastructure have limited fisheries, maritime transport, tourism, investment, and other blue economy activities in Rivers State. The study concludes that strengthening of maritime security, improving governance, protecting the marine environment and investing in maritime infrastructure can unlock the full potential of the blue economy as a sustainable driver of economic diversification in Rivers State.

Key words: Blue economy, Insecurity, Rivers state, Marine Economy

Introduction

Few states in Nigeria are as closely connected to water as Rivers State. Across its creeks, rivers, and coastal settlements, generations of people earned a living from fishing, water transportation, and trade. Long before petroleum became the dominant feature of the state's economy, waterways connected communities and supported local commerce. Even after the growth of the oil industry, many riverine communities continued to depend on aquatic resources for their livelihood (Fubara, 1998; Igali, 2024).

The renewed interest in the blue economy has drawn attention to this older relationship between people and water. Much of the discussion centres on activities such as fishing, aquaculture, maritime transport, shipping, tourism, and port services (Jacob & Umoh, 2022; Oti et al., 2024). Rivers State possesses the natural setting for these activities. Its waterways stretch across much of the state, while its coastal location provides access to maritime trade and other economic opportunities linked to the sea.

Yet there is another side to this picture. In many parts of the state, waterways that sustain

economic activity have also become sites of insecurity. Reports of piracy, kidnapping, sea robbery, illegal oil bunkering, and cult-related violence are not uncommon. For those who depend on the waterways, these developments affect everyday decisions about movement, work, and investment. In some communities, insecurity has become part of the conditions under which economic life is conducted (Ezeani, 2013; Ukeje & Wanjohi Ela, 2013). Environmental degradation presents a related challenge. Oil spills and illegal refining have affected rivers and creeks in different parts of the Niger Delta. Communities that rely on fishing often feel the consequences directly through declining catches and reduced income. The condition of the aquatic environment therefore has implications not only for ecology but also for livelihood (Kanu & Archibong, 2011; Ordinioha & Brisibe, 2013)

The growing attention given to the blue economy makes these issues particularly important. Rivers State possesses resources that could support employment, trade, and income generation, but the benefits derived from those resources depend largely on the circumstances in which they are used (Jacob & Umoh, 2022; Oti et al., 2024). Although insecurity has received considerable attention in studies of the Niger Delta, less attention has been given to its implications for the utilisation of marine resources in Rivers State. This study examines that relationship and considers what it reveals about the prospects of blue economy development in the state.

The study provides a historical overview of Nigeria's economy, examines the potential of the blue economy in Rivers State, investigates the security challenges affecting its development, and assesses the socioeconomic and environmental impacts of those challenges on blue economy activities in the state.

Human Security Theory as a Framework for Understanding Insecurity and the Blue Economy

This study adopts Human Security Theory as its analytical framework. Popularized through the United Nations Development Programme's Human Development Report (1994), the theory challenged the traditional tendency to view security mainly in terms of military defence and territorial protection. Mahbub ul Haq argued that security should also address threats that affect people's daily lives and livelihoods. Consequently, the scope of security was broadened to include economic, food, health, environmental, personal, community and political security (United Nations Development Programme [UNDP], 1994).

Human Security Theory is people-centred and multidisciplinary. It proceeds from the view that the well-being of individuals and communities is as important as the security of the state. Security is therefore not limited to the protection of borders but extends to the protection of livelihoods, basic needs and the conditions necessary for people to live productive and dignified lives (James, 2014; Lautensach & Lautensach, 2020; Pitsuwan & Caballero-Anthony, 2014). The theory further maintains that development is difficult to sustain where insecurity restricts access to opportunities and productive resources (Tadzbakhsh & Chenoy, 2007).

The relevance of the theory to this study lies in the close relationship between security and the blue economy. Rivers State possesses extensive marine and aquatic resources that support fishing, maritime transportation, tourism and trade. The benefits associated with these activities, however, depend largely on the conditions under which they are carried out. Piracy, kidnapping,

sea robbery, cult-related violence, illegal oil bunkering, oil theft and environmental pollution continue to affect many waterways and coastal communities in the state (Ezeani, 2013; Joab-Peterside et al., 2021; Ukeje & Wanjohi Ela, 2013). These challenges disrupt livelihoods, discourage investment and reduce the economic value that can be derived from marine resources.

Human Security Theory helps explain this situation by drawing attention to the consequences of insecurity for human welfare and economic activity. Where individuals are unable to safely utilize available resources, development is constrained regardless of the abundance of those resources. The theory is therefore useful for examining how insecurity affects the utilisation of marine resources and limits the growth of the blue economy in Rivers State.

Literature Review

Aladejana et al. (2024) considered the impact of blue economy on sustainable development in Nigeria using annual time series data from 1990 to 2021. Sources: Central Bank of Nigeria Statistical Bulletin, and World Bank Development Indicators. Descriptive statistics, correlation analysis, Augmented Dickey-Fuller unit root tests, Johansen cointegration tests and the Ordinary Least Square (OLS) estimation technique were employed in the study. The results show that blue economy has positive significant effect on sustainable development outcomes whereas CO2 emissions and per capita income growth have negative significant effect on sustainable development outcomes. The authors stated that for Nigeria to achieve sustainable development, integration of blue economy initiatives into national development planning through policies that promote economic growth, environmental sustainability and social inclusion is a pre-requisite.

Jacob and Umoh (2022) therefore examined the opportunities, problems and challenges associated with the development of the blue economy in Nigeria by reviewing the literature, policy documents and international framework on marine resource governance. The blue economy encompasses vital components such as fisheries, maritime transport, tourism, offshore energy resources and marine biodiversity that can drive economic diversification. But they said the sector still faces challenges of maritime insecurity, environmental degradation, weak institutional frameworks, inadequate infrastructure, climate change, pollution and limited technology capacity. Aladejana et al. (2024) emphasised the developmental aspect of Blue Economy at the national level, while Jacob and Umoh (2022) emphasised the institutional and security issues that hinder the achievement of the potential. The two studies both suggest that to realise the developmental dividends of the blue economy, effective governance, environmental protection and maritime security, as well as the availability of resources, are needed.

Tamunodukobipi and Iheakam (2023) explored the ports and maritime sector for the engineering infrastructure for the development of Nigeria's blue economy. The study used documentary evidence of cargo traffic, port performance and infrastructure indicators for a period of fifteen years. It said "maritime development constraints remain inadequate deep seaports, poorly dredged navigation channels, old cargo handling equipment, weak road and rail connectivity, limited ship building facilities and inadequate maritime security infrastructure". These gaps are leading to congestion in ports, high logistics costs, long turnaround times for vessels, less efficient trade and underutilised maritime resources. The study agrees with the

findings of Jacob and Umoh (2022) that lack of infrastructure is another obstacle to harnessing Nigeria's blue economy potential.

Amos et al. (2024) studied the relationship between blue economy and sustainable use of water resources for economic development of coastal communities in Calabar, Cross River State. Survey design, questionnaires, structured interviews and Pearson Product Moment Correlation were used to establish significant positive relationships between sustainable fishing, sustainable periwinkle harvesting and blue economy development. The authors concluded that effective fisheries and aquatic resource management enhances household livelihoods, promotes local economic development and contributes to the long-term sustainability of coastal ecosystems. The result corroborates the findings of Aladejana et al. (2024) and Jacob and Umoh (2022) on the crucial role of sustainable management of marine resources for economic growth and environmental sustainability.

For instance, Oti, George and Amawei (2024) in their study explored the impact of the blue economy for industrial development in Rivers State, Port Harcourt Metropolis through questionnaires, interviews and focus group discussions. The study found that fisheries, maritime transport and tourism were the main drivers of industrial growth. It also found that the development of the industrial sector is positively influenced by infrastructure development, policy reforms and sustainable resource management while environmental degradation, poor governance, weak implementation of regulations, inadequate infrastructure and poor coordination of stakeholders limit the development of the sector. Unlike the broader national studies, Oti et al. (2024) provides empirical evidence from Rivers State and shows the direct impact of local governance and infrastructure on blue economy development.

The body of academic literature on the link between insecurity and the blue economy is expanding. Asua and Nwangwu (2025) investigated the effects of maritime insecurity and environmental degradation on the development of Nigeria's blue economy with a special focus on the Niger Delta. Focus group discussions, key informant interviews, field observations and documentary evidence showed that armed robbery at sea, artisanal crude oil refining, oil spills and unsustainable fishing practices have degraded marine ecosystems, disrupted maritime trade, reduced fish stocks and weakened the livelihoods of coastal communities. These challenges have also been aggravated by lack of maritime security, poor governance, weak environmental regulation and limited community participation, the study said. Their findings corroborate the position of Jacob and Umoh (2022) that good governance and maritime security are vital to the sustainable development of the blue economy.

Jack Joab-Peterside Nyiayaana (2021) revealed that conflict economy driven by oil bunkering, artisanal refining, piracy, kidnapping, cultism and political patronage impoverished fisheries, maritime transport and other marine based livelihoods through pollution and insecurity. Similarly, Dagogo (2022) said illegal oil bunkering has triggered armed violence, kidnapping and organised criminality, damaged aquatic ecosystems, discouraged legitimate investment, reduced government revenue and impeded sustainable development. Both studies focus on insecurity in Rivers State, but while Joab-Peterside et al. (2021) take a wide governance perspective, Dagogo (2022) focuses specifically on illegal oil bunkering as a major driver of insecurity. Together, they show that insecurity affects not only public safety but also the sustainability and productivity of marine-based economic activities.

The reviewed literatures generally agreed that blue economy has great potential to promote economic diversification, employment creation, industrial development and sustainable growth in Nigeria. There is also broad consensus that maritime insecurity, environmental degradation, weak governance, lack of infrastructure and poor regulatory enforcement continue to hinder the realisation of these opportunities. However, most of the research carried out so far has adopted a mainly national approach to the blue economy or has considered insecurity as one of the factors influencing development at sea. There is relatively less attention on a historical account on the impact of insecurity on the trajectory of the blue economy in Rivers State; incorporating its security, socio-economic and environmental dimensions. This study has filled this gap by providing the historical background of Nigeria's economy; analysing the security threats to the development of the blue economy in Rivers State; examining the security threats to the development of the blue economy in Rivers State and assessing the socio-economic and environmental impacts of the threats on blue economy activities in the state.

Historical Overview of Nigeria's Economy

Before crude oil was discovered, Nigeria's economy thrived on a vibrant and varied mix of agriculture, trade, fishing, mining and local industries. Agriculture was the mainstay of the economy during the colonial period and the early post-independence years, generating most of the country's export earnings. The basis of external trade was the cash crops such as cocoa from the Western Region, groundnuts from the Northern Region and palm produce from the Eastern Region. Food crop cultivation supplied the domestic food stock. Agriculture, fishing and stock raising, mining and indigenous crafts such as weaving, pottery, blacksmithing, leatherwork, textile making and wood carving were important sources of employment, local trade and economic growth. The economy also benefited from regional and long-distance trade as markets linked the various ecological zones. The exchange of agricultural products, manufactured goods, salt, textiles, livestock and other merchandise was facilitated through the Trans-Saharan trade in the north and coastal trade in the south. These commercial networks encouraged economic specialisation, stimulated commercial activities and promoted the growth of important trading centres throughout the country (Afigbo, 1987; Falola & Heaton, 2008; Hopkins, 1973).

The discovery of crude petroleum in commercial quantity at Oloibiri in present-day Bayelsa State in 1956 and the commencement of commercial production in 1958 gradually transformed Nigeria's economic structure. Although agriculture remained an important source of revenue in the years immediately following independence, the rapid expansion of petroleum production after the Nigerian Civil War and the oil boom of the 1970s fundamentally altered the country's economic priorities. Increased oil revenue redirected government expenditure and public investment towards the petroleum industry, while agriculture, manufacturing, mining and many traditional productive sectors received relatively less policy attention. By the late 1970s, Nigeria's economy had become increasingly dependent on crude oil as its principal source of foreign exchange earnings and government revenue (Umar, 2019).

The concentration of economic activities in the petroleum sector exposed the country to fluctuations in the international oil market. Periodic declines in global crude oil prices resulted in falling government revenues, foreign exchange shortages, fiscal constraints and slower economic growth. These developments demonstrated the vulnerability of Nigeria's mono-

product economy and reinforced the need to diversify the country's productive base. Consequently, successive governments introduced different policies and programmes aimed at revitalising the non-oil sectors to promote sustainable economic growth, create employment opportunities and expand revenue generation (Umar, 2019).

As part of this broader economic diversification agenda, increasing attention has been given to the blue economy. Nigeria's extensive coastline, inland waterways, rivers, creeks, estuaries and marine resources provide enormous opportunities for fisheries, aquaculture, maritime transportation, shipping, port operations, coastal tourism, marine biotechnology, offshore renewable energy and other water-based industries (Jacob & Umoh, 2022; Aladejana et al., 2024). For centuries, these aquatic resources have supported livelihoods, commerce and transportation in many coastal communities, although their economic potential remained relatively underutilised during the period of heavy dependence on petroleum (Jacob & Umoh, 2022).

Recent policy initiatives, particularly the establishment of the Federal Ministry of Marine and Blue Economy in 2023, reflect a growing recognition of the maritime sector as an important component of Nigeria's economic diversification strategy. The blue economy is increasingly viewed as a means of creating employment, expanding maritime trade, increasing government revenue, promoting industrial development and advancing sustainable economic growth (Aladejana et al., 2024; Jacob & Umoh, 2022). The increasing significance of the blue economy can therefore best be understood within the broader historical evolution of Nigeria's economy from an agrarian system to an oil-dependent economy and, more recently, towards economic diversification through the sustainable utilisation of marine and aquatic resources (Aladejana et al., 2024; Jacob & Umoh, 2022).

The Potential of the Blue Economy in Rivers State

Rivers State is strategically positioned in the Nigerian maritime domain and is endowed with some of the country's richest marine and aquatic resources. For centuries, the vast array of rivers, creeks, estuaries, mangrove forests and Atlantic coastline have influenced settlement patterns, livelihoods and trade. Long before petroleum became the mainstay of the state's economy, fishing, canoe transportation, boat building and riverine trade sustained many coastal and inland communities. These natural endowments still provide a good basis for the development of the blue economy and put Rivers State on the path of diversifying its economy away from crude oil (Fubara, 1998; Jacob & Umoh, 2022; Igali, 2024).

Fisheries and aquaculture are among the most important components of the blue economy. Rivers State is endowed with a large number of fish species, shrimps, oysters, crabs, periwinkles and other aquatic resources in its rivers, estuaries, creeks and coastal waters which have sustained the riverine communities for centuries. Artisanal fishing is still an important occupation in communities like Andoni, Bonny, Opobo, Kalabari and Degema while commercial aquaculture has continued to grow in different parts of the state. Besides providing food and jobs, these activities provide opportunities for processing, preserving, packaging, transporting, marketing and exporting fish. Thus fisheries and aquaculture have the potential to stimulate agro-processing industries, extend value chains, increase household incomes, reduce dependence on imported fish and generate further revenue for the state (Anyanwu, 2014; Amos

et al., 2024; Jacob & Umoh, 2022). As highlighted by Amos et al. (2024), the sustainable management of fisheries and aquatic resources is significant for household livelihoods, economic growth and the long-term preservation of coastal ecosystems. Similarly, Jacob and Umoh (2022) identified fisheries and aquaculture as the key sectors for the blue economy to promote economic diversification in Nigeria.

Maritime transport is another sector that can change the state's economy. Historically, rivers and creeks have been important modes of transportation connecting riverine settlements and enabling trade in the Niger Delta (Igali, 2024). Rivers State today hosts two of Nigeria's major seaports, the Port Harcourt Port and the Onne Port, which handle international trade, oil and gas logistics, container traffic and industrial cargo. Better port infrastructure, dredged navigation channels, strengthened cargo-handling facilities and better integration of road and rail transport would improve port efficiency, attract private investment, reduce logistics costs, increase customs and port revenues and promote industrial expansion (Tamunodukobipi & Iheakam, 2023; Oti et al., 2024). As maritime trade continues to grow, Rivers State is well positioned to consolidate its position as one of Nigeria's leading commercial and logistics hubs. Oti et al. (2024) found that maritime transport was among the sectors with significant capacity to boost industrial growth within the state.

Tourism is another area where the blue economy can drive economic transformation. Rivers State has many beaches, islands, mangrove landscapes, rivers and historic settlements that provide opportunities for beach tourism, ecotourism, cultural tourism and recreational activities. Bonny, Opobo, Degema, Akuku-Toru, Kono and other coastal riverine settlements possess natural and cultural attractions that can attract both domestic and international visitors. Investment in hotels, resorts, water transport, restaurants, recreational facilities, tour operations and cultural festivals will not only generate revenue but will also stimulate small and medium-scale enterprises, encourage local entrepreneurship and create employment across the hospitality, transport, entertainment and creative sectors. Through these multiplier effects, tourism can be a major contributor to the non-oil economy of the state (Fubara, 1998; Oti et al., 2024).

Rivers State is also blessed with rich biodiversity which further enhances the potential of its blue economy. The state's mangrove forests, estuaries, rivers and coastal waters provide habitat for a variety of fish, crustaceans, marine mammals, reptiles and migratory birds. The richness of these water ecosystems and their economic importance are evidenced by Oyorokoto and other coastal communities. Sustainable management and conservation of marine biodiversity can support fisheries, scientific research, environmental education and ecotourism while ensuring that these resources will continue to generate economic benefits for future generations (Amos et al., 2024; Jacob & Umoh, 2022).

With these traditional sectors, the blue economy offers opportunities in emerging marine industries. Marine biotechnology has potential for pharmaceutical research, industrial biotechnology and development of products from marine organisms. Offshore renewable energy, such as offshore wind and tidal energy, offers alternative clean energy sources that can assist in fostering sustainable industrial development. Rivers State also provides a conducive environment for marine engineering services, shipbuilding, ship repairs, maritime logistics, offshore support services and environmentally friendly exploration of seabed resources. Many

of these sectors are still in their infancy, but they have the potential to attract investment, promote technological innovation, build skilled manpower and strengthen the state's industrial capacity (Jacob & Umoh, 2022; Aladejana et al., 2024).

Rivers State's strategic location in the Gulf of Guinea further enhances these opportunities. Shipping and port services play a crucial role in world trade. A large proportion of global trade is transported by sea (World Economic Forum, 2024). Rivers State's proximity to the coast and existing maritime infrastructure provide the state with a favourable position to take advantage of regional and global trade. Improved shipping services, efficient port operations, modern logistics systems and increased maritime commerce would enhance commercial activities, generate more revenue and entrench the state's position as one of Nigeria's leading maritime gateways (Oti et al., 2024; Tamunodukobipi & Iheakam, 2023).

Security Threats to the Blue Economy in Rivers State

The blue economy of Rivers State is strongly connected to the security of its maritime environment. With its vast network of rivers, creeks, estuaries, mangrove forests and Atlantic coastline, Rivers State's strategic location in Nigeria's maritime domain is well established. These geographical features offer favourable conditions for fisheries, aquaculture, maritime transport, tourism, port operations and other marine-based economic activities. However, the development of these sectors has faced a cluster of security challenges that have evolved over time. These are militancy, piracy, illegal oil bunkering, kidnapping, cultism, environmental degradation, weak maritime governance and poor maritime infrastructure. The combination of these challenges has shaped the trajectory of the blue economy in the state.

These security challenges occurred in the process of the historical evolution of the oil economy of the Niger Delta. Until the commercial exploitation of Petroleum, most of the riverine communities in the present day Rivers State depended on fishing, canoe transportation, local trade and other aquatic occupations for their livelihood. After the discovery and commercial production of crude oil, Rivers State became one of the major petroleum-producing regions in Nigeria. Notwithstanding the immense wealth generated from oil, many host communities were still faced with environmental pollution, unemployment, poor infrastructure and limited participation in the management of petroleum resources. Discontent resulting from these conditions led to the rise of resource control movements and militant groups in the late 1990s and early 2000s. These groups initially emerged as environmental justice and resource control advocates, but the security situation has since become more intricate due to the emergence of numerous armed groups in the waterways of Rivers State (Aladejana et al., 2024; Jacob & Umoh, 2022; T. Batubu, personal communication, August 4, 2024).

The blue economy in Rivers State is riddled with several security challenges, one of which is militancy. During the early years of the Fourth Republic, militant groups attacked oil installations, flow stations, pipelines and oil service facilities in the Niger Delta. Rivers State has become a major hub for such activities because of its concentration of petroleum infrastructure and multinational oil companies. Although the Federal Government's Amnesty Programme of 2009 reduced the organised militant confrontations in the maritime environment, it did not completely eradicate the insecurity in the maritime environment. Instead, the security environment shifted progressively from organised militant campaigns to decentralised criminal

networks engaged in piracy, kidnapping, illegal oil bunkering and other maritime crimes (Aladejana et al., 2024; Ezeani, 2013).

Piracy is another serious security challenge for the blue economy in Rivers State. Piracy has been around on the seas for as long as there has been history and there are records of it from the old Mediterranean world. But today the Gulf of Guinea has become one of the world's leading hotspots for piracy. A significant number of piracy incidents recorded in the region take place in Nigeria's territorial waters, especially within those around the Niger Delta (United Nations Office on Drugs and Crime [UNODC], 2022; Ukeje & Wanjohi Ela, 2013). Rivers State is at the heart of this maritime zone, with numerous creeks and waterways giving access to international shipping routes as well as offshore petroleum facilities. Piracy is one of the major maritime security concerns in the state. Pirate groups in these waters have attacked fishing boats, passenger ships, offshore support vessels, cargo ships and oil service craft (Ezeani, 2013).

The experiences of Oyorokoto fishing settlement in Andoni Local Government Area paint a picture of the persistence of piracy in Rivers State. Oyorokoto is one of the largest fishing settlements in West Africa with substantial marine resources. But the fishermen community has many times reported attacks by armed pirates along the waterways, despite their strategic importance. The recurrence of these events shows that organised maritime criminal groups still exist in the coastal environment of Rivers state (Today FM, 2024). Similarly, the commercial value of a dolphin captured in Oyorokoto in 2020 and the public attention it generated, further underscores the strategic importance of the state's marine resources and the need for improved maritime security (BBC News Pidgin, 2020).

Illegal oil bunkering is another serious threat to security in the maritime environment of Rivers State. The extensive network of creeks and rivers has provided a conducive environment for organised criminal groups to establish illegal refining camps and crude oil theft operations in many riverine communities. Illegal bunkering has developed into a sophisticated criminal activity, including the sabotage of pipelines, the extraction of crude oil, and the transportation of stolen petroleum products via inland waterways. Research on Nigeria's maritime security underscores illegal oil bunkering as one of the major security concerns in the Niger Delta due to its close association to organised crime and maritime criminality (Ezeani, 2013; Ukeje & Wanjohi Ela, 2013).

Kidnapping has also been a major security challenge in Rivers State. Initially, kidnapping was associated with militant attacks on foreign oil workers but it later extended to politicians, businessmen, professionals, traders and ordinary citizens. Criminal groups have often exploited riverine communities and waterways as operational routes that afford relatively hidden access between coastal and inland communities. The prevalence of kidnapping is a manifestation of the general insecurity that has plagued many parts of Rivers State since the early 2000s (Aladejana et al., 2024).

Cultism is another major dimension of insecurity in Rivers State. Cult groups first appeared in tertiary institutions, but many later evolved into organised armed groups in urban and riverine communities. Increasingly, these organisations have been associated with violent confrontations, territorial disputes, armed robbery, political violence and organised criminal activities. (E. Nwei, personal communication, September 14, 2024) Community security

personnel in Rivers State identified cult-related violence as one of the persistent security challenges faced by some communities in Rivers State. The persistent cult activities highlight the complexities of the state's internal security landscape, going beyond the scope of conventional maritime crimes.

Environmental degradation has also emerged as a security challenge in the blue economy. Oil exploration and production activities have led to recurrent oil spill incidents in Rivers State over several decades. Communities like Bodo, Okrika and Umuechem have been subjected to serious environmental pollution due to oil production. It is estimated that on average 240,000 barrels of crude oil are spilt every year within the Niger Delta, polluting rivers, creeks and wetlands (Nigerian Medical Journal, 2013). In August 2023, an oil spill in Okrika, which saw large quantities of dead fish floating on surrounding waters, drew national attention to the extent of aquatic pollution in the state (Naku, 2023). These events highlight the persistent environmental problems facing the aquatic ecosystems of Rivers State.

Apart from the petroleum pollution, another threat to the environmental security is the discharge of industrial effluents into the rivers and creeks. Breweries, chemical factories, pharmaceutical companies, detergent makers and palm oil processors often dump their waste in water bodies where the treatment facilities are poor. These industrial discharges are contributing to the deterioration of water quality and increasing the ecological stress on the aquatic environments of Rivers State (Kanu & Archibong, 2011).

The security environment is further complicated by weak maritime governance and weak regulatory enforcement. Previous studies on Nigeria's blue economy have identified poor institutional coordination, weak enforcement of maritime laws and inadequate policy implementation as ongoing governance challenges affecting the sustainable management of marine resources (Aladejana et al., 2024; Oti et al., 2024). The study in Port Harcourt also points to governance, regulatory enforcement, and stakeholder coordination as crucial issues requiring constant attention within the blue economy framework in Rivers State (Oti et al., 2024).

Another big security challenge has to do with maritime infrastructure. Rivers State has two major seaports, Onne Port and the Port Harcourt Port Complex, and numerous jetties and navigable waterways. However, interviews with maritime stakeholders indicate concerns about insufficient dredging of navigation channels, low port utilisation and navigational constraints affecting larger vessels (I. Ugo, personal communication, 2024; M. Okechi, personal communication, 2024; E. Frank, personal communication, 2024). Similarly, the 2024 blue economy study in Port Harcourt identifies infrastructural deficiencies as one of the major challenges of maritime development in Rivers State (Oti et al., 2024).

Impact of Security Challenges on the Blue Economy in Rivers State

The blue economy has the capacity to contribute significantly to the diversification of the Rivers State economy through fisheries, aquaculture, maritime transportation, port operations, coastal tourism, marine services and other ocean-based industries. The State's long coastline, numerous rivers, creeks, estuaries and strategic maritime position provide conducive conditions for sustainable economic activities. But the continued security challenges have greatly limited the ability of Rivers State to tap the full potential of these marine resources. The performance of the

state's blue economy has been impacted by the confluence of militant activities, piracy, illegal oil bunkering, kidnapping, cultism, environmental pollution, weak maritime governance, and infrastructural deficiencies (Aladejana et al., 2024; Oti et al., 2024). These challenges have deeply affected the socioeconomic and environmental aspects of virtually all sectors of the maritime economy.

One of the most important effects has been on fisheries and aquaculture. Fishing is still one of the oldest occupations of the riverine communities of Rivers State, particularly in Andoni, Bonny, Akuku Toru, Asari Toru and Degema local government areas. The rising incidence of piracy, sea robbery and kidnapping has made much of the fishing grounds unsafe. The consequence has been that many artisanal fishermen have reduced the number of fishing trips or stopped fishing in the deep sea altogether because they fear attacks on the waterways. This has led to reduced seafood production and household income of fishing communities and decline in fish catch. The experiences of fishermen in Oyorokoto Fishing Settlement, one of the largest fishing settlements in West Africa, exemplify this challenge. Fishermen in the settlement have repeatedly reported attacks by armed pirates operating in the waterways, disrupting fishing activities and threatening livelihoods (Today FM, 2024). This is contrary to the principles of the blue economy that advocate for sustainable use of fisheries as a source of food security, employment and economic development (Amos et al., 2024).

Security concerns have also limited investment in fisheries, aquaculture and other marine industries. The continuing insecurity and weak protection offered by institutions are generally not encouraging to investors to establish businesses. Consequently, opportunities for expansion of commercial aquaculture, seafood processing, fish preservation and other marine enterprises have stayed below their potential. Recent studies on Nigeria's blue economy point to the need for effective governance, secure maritime environments, and supportive investment policies for sustainable maritime development (Aladejana et al., 2024). Similarly, the Port Harcourt study identifies insecurity, governance weaknesses, and infrastructural deficiencies as major constraints limiting industrial growth within the blue economy of Rivers State (Oti et al., 2024).

Another important effect is the underdevelopment of tourism along the coast. Rivers State boasts of many coastal attractions that can support tourism like Akuku Toru Beach, Degema Beach, Kono Beach, Bonny Beach, Opobo Beach, Ikuru Beach and other historical coastal communities. Oyorokoto Fishing Settlement also has important ecological resources including dolphins, sea turtles and the nearby Ikuru Elephant Reserve, all of which offer opportunities for eco-tourism. The capture of a dolphin in Oyorokoto in 2020 drew wide public attention and showed the richness of the state marine biodiversity (BBC News Pidgin, 2020). However, insecurity has prevented many of these attractions from reaching their tourism potential. Kidnapping, piracy, cult violence and other criminal activities are common, discouraging many coastal destinations from receiving domestic and international tourists. Local evidence from Borokiri also suggests that security concerns have negatively impacted the growth and attractiveness of the Port Harcourt Tourist Beach and other recreational centres in the state (A. Umejuru, personal communication, September 10, 2024). This has restricted private investment in tourism infrastructure and hampered the contribution of tourism to the blue economy of the state.

The same applies to maritime transportation and port operations that have suffered from

persistent insecurity. Rivers State plays a strategic role in Nigeria's maritime economy through the Port Harcourt Port Complex and Onne Port. Maritime transport continues to be important to international trade, with around 90 per cent of world trade by volume shipped by sea (World Economic Forum, 2024). However, operational risks for vessels transiting Nigerian waters have increased, with piracy and armed attacks in the Gulf of Guinea. Shipping companies are forced to spend more on insurance, private security and other risk management measures thus increasing the total cost of maritime transportation (Ukeje & Wanjohi Ela, 2013; Ezeani, 2013). The extra costs make maritime business less competitive in the state and reduce the incentive to use its ports more.

Insecurity and lack of maritime infrastructure have combined to contribute to under-utilisation of the state's port facilities. Rivers State has two major seaports: The Port Harcourt Port and the Onne Port. But interviews with importers and maritime professionals show that many businesses prefer to clear their cargo through Lagos, Cotonou and Lomé due to lower operational costs and fewer logistical constraints. In addition, poor dredging of navigation channels, navigational constraints for bigger vessels and inefficient port operations were also mentioned by respondents as other challenges constraining maritime commerce in Rivers State (M. Okechi, personal communication, 2024; E. Frank, personal communication, 2024; I. Ugo, personal communication, 2024). These findings are in agreement with the findings of Oti et al. (2024) who identified infrastructural deficits as one of the major hindrances to the growth of the blue economy in Port Harcourt.

Another major impact of insecurity has been the reduction of investor confidence and industrial development in the maritime sector. The blue economy's growth depends on sustained investments in shipping, fisheries, marine transport, tourism, renewable energy and other ocean-based industries. However, the rise of piracy, kidnapping, illegal oil bunkering and cult-related violence have increased business risks and discouraged both local and foreign investment. The situation has led to the slow death of industrial activities in some parts of Rivers State, including Trans-Amadi, where insecurity has discouraged business expansion and industrial investment. This perception of Rivers State as a high risk destination for investment has continued to limit the diversification aspiration of the blue economy.

Security challenges have also led to unemployment and declining household incomes in coastal communities. A large number of households in the riverine areas depend directly or indirectly on fishing, fish processing, boat transportation, marine trading and tourism related activities. These occupations are disturbed by insecurity which results in lower household incomes and higher levels of unemployment, especially among young people. In some cases, the lack of legitimate economic opportunities has led to more illicit activities, including piracy, crude oil theft and illegal artisanal refining, which further fuels the cycle of insecurity in the maritime environment.

The environmental consequences of insecurity have been just as serious. Rivers State has been hit by rampant oil bunkering, pipeline vandalism, artisanal refining and recurrent oil spills which have led to widespread pollution of rivers, creeks and wetlands. Communities such as Bodo, Okrika and Umuechem have suffered repeated environmental degradation from petroleum exploration and transportation. Available evidence estimates that about 240,000 barrels of crude oil are spilt annually in the Niger Delta, leading to widespread contamination of

the aquatic ecosystem and biodiversity (Ordinioha & Brisibe, 2013). The ecological damage to aquatic resources was shown when in August 2023, an oil spill in Okrika drew national attention as residents gathered dead fish floating on polluted waters (Naku, 2023). Such environmental degradation is contrary to one of the main aims of the blue economy, which is to reconcile economic use with environmental sustainability.

Environmental degradation has been worsened by industrial pollution in Rivers State. Breweries, chemical industries, pharmaceutical companies, detergent manufacturers, palm oil processing industries still pollute rivers and creeks with untreated effluents. Pollutants impact water quality and endanger aquatic biodiversity, consequently diminishing the ecological resilience of marine ecosystems that underpin fisheries, tourism and other blue economy sectors (Kanu & Archibong, 2011). More recent scholarship on the blue economy has continually emphasised the significance of healthy aquatic ecosystems for sustainable marine development (Amos et al., 2024; Aladejana et al., 2024).

Conclusion

Rivers state stands at a critical juncture in its economic history, with the need for diversification more pressing than ever. The blue economy offers a viable alternative to the country's over-reliance on oil, providing opportunities for sustainable economic growth. However, the development of the blue economy is contingent on addressing the security and environmental challenges that have hindered its progress.

This research has shown that with the right measures in place, the blue economy could play a pivotal role in Nigeria's future economic strategy. Rivers State, in particular, stands to benefit immensely from the development of its maritime resources. To achieve this, stakeholders at all levels must prioritize the security and sustainability of Nigeria's waterways as insecurity and the underdevelopment of blue economy are intrinsically tied together, ensuring that the blue economy becomes a cornerstone of the nation's economic diversification efforts.

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